

PROJECT DESIGN PHASE : PROBLEM - SOLUTION FIT CANVAS

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

Date: 31 July 2026

Team / Dev ID: 23JR1A05E4 (Individual)

Developer Name: Shaik Zuveriya Tabassum

Max Marks: 2 Marks

1 CUSTOMER SEGMENT(S) [CS] Daily Commuters: Office workers & students needing fast ticket booking during rush hours. Casual Travelers: Occasional passengers seeking direct ticket generation without app registration. Station Staff: Inspectors validating QR tickets and managing turnstile flow.	2 JOBS-TO-BE-DONE / PROBLEMS [PB] Generate Tickets Instantly: Fast, digital self-service booking via mobile. Accurate Fare Preview: Automatic calculation based on dynamic source-destination matrix. Turnstile Access: Scannable QR code generation for entry/exit validation.	3 TRIGGERS [TR] - Arriving at packed metro stations with long queue lines at token counters. - Running late for scheduled train departure. - Misplacing physical plastic transit tokens during journey.	4 EMOTIONS (BEFORE / AFTER) [EM] Before: Frustrated, anxious about missing trains in queue lines. After: Relieved, confident, satisfied with seamless QR access.
5 AVAILABLE SOLUTIONS [AS] Physical Counters / Kiosks: Paper tokens/cards causing long queue bottlenecks. Third-Party Apps: Slow sync with station systems and lacks direct validation logic. Pros/Cons: Physical counters are simple but cause heavy rush-hour delays.	6 CUSTOMER CONSTRAINTS [CC] Network Connectivity: Spotty cellular signals inside underground stations. Time Limitations: Ticketing must take < 30 seconds during peak commute times. Usability: UI must be simple for non-tech-savvy passengers.	7 BEHAVIOUR [BH] - Commuters prefer using smartphones while walking toward station entry gates. - Strong user shift toward contactless digital payments and instant web-portal access.	8 CHANNELS OF BEHAVIOUR [CH] Online Channels: ServiceNow Service Portal, automated Email/SMS notifications, REST integrations. Offline Channels: Station QR signage, turnstile scanners, inspector handheld mobile devices.
9 PROBLEM ROOT CAUSE [RC] Manual Overhaul: Reliance on human ticket vendors creates heavy station congestion. Disconnected Tools: Lack of real-time unified engine linking fare, ticket lifecycle, and scan validation.		10 YOUR SOLUTION [SL] ServiceNow Service Portal: Responsive self-service portal for quick booking. GlideAjax Fare Engine: Dynamic distance matrix fare calculation in seconds. QR Code Engine & Workflows: Instant QR generation and automated status transitions (Active → Used/Expired). Inspector UI: Mobile portal for inspectors to scan and validate tickets on duty.	

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